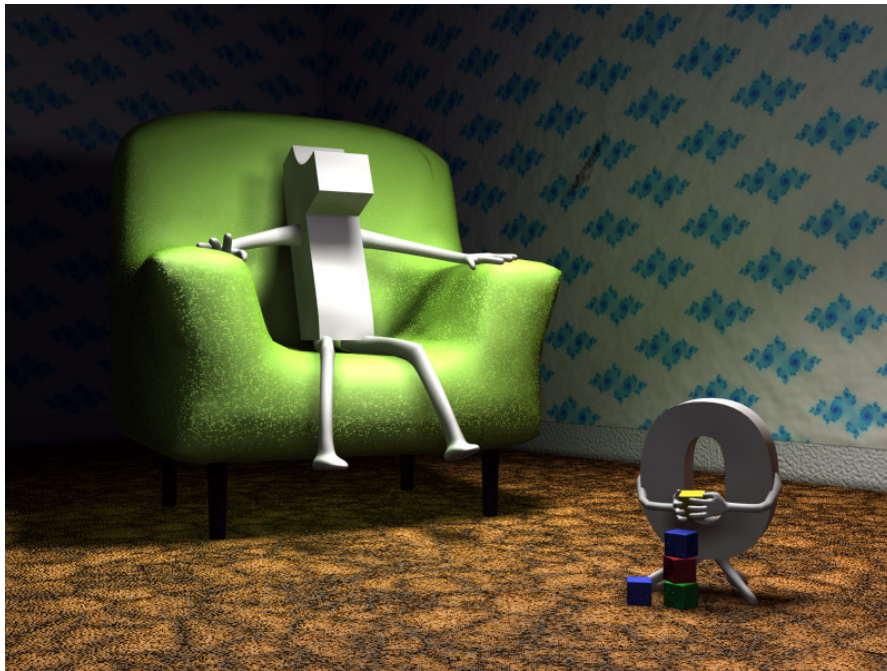


Chapter 1. INTEGERS

1. 3 Integers modulo n

1.3.2 Solving linear equations and systems in $\mathbb{Z}_n$



Today we will learn how to solve linear equations and linear systems using modular arithmetic.

Equations we will investigate today:

$$\bar{a}x = \bar{c} \text{ in } \mathbb{Z}_n \Leftrightarrow ax \equiv c \pmod{n}$$

$$\left. \begin{array}{l} \bar{a}x + \bar{b}y = \bar{r} \\ \bar{c}x + \bar{d}y = \bar{s} \end{array} \right\} \text{ in } \mathbb{Z}_n \Leftrightarrow \begin{cases} ax + by \equiv r \pmod{n} \\ cx + dy \equiv s \pmod{n} \end{cases}$$

Definition 1.44. Let $n \geq 2$ and a, b be integers. A **residue class** $\bar{b} \in \mathbb{Z}_n$ is called an **inverse** of $\bar{a}$ if $\bar{a}\bar{b} = \bar{b}\bar{a} = \bar{1}$ in $\mathbb{Z}_n$.

THEOREM 1.45. Let a and n be integers with $n \geq 2$. Then $\bar{a}$ has an inverse in $\mathbb{Z}_n$ if and only if a and n are relatively prime. Moreover, the inverse of $\bar{a}$ in $\mathbb{Z}_n$ is the residue class of the coefficient of a in Bézout's Lemma.

Proof. $\bar{b}\bar{a} = \bar{1} \text{ in } \mathbb{Z}_n \Leftrightarrow ba = 1 \text{ in } \mathbb{Z}_n$

$$\Leftrightarrow ba \equiv 1 \pmod{n} \Leftrightarrow n \mid (ba - 1)$$

$$\Leftrightarrow ba - 1 = nq \text{ for some integer } q.$$

$$\Leftrightarrow 1 = \underline{ba} + (-q)n \Leftrightarrow \gcd(a, n) = 1.$$

$\uparrow$
 Bézout's Lemma



Example 1.46. Find the inverse of $\overline{16}$ in $\mathbb{Z}_{35}$.

$$\gcd(16, 35) = 1.$$

We apply the Euclidean Algorithm:

$$35 = 2 \cdot 16 + 3 \quad (1)$$

$$16 = 5 \cdot 3 + \underline{1} \quad (2)$$

$$3 = 3 \cdot 1 + \underline{0}$$

$$\underline{1} \stackrel{(2)}{=} 16 + (-5) \cdot \underline{3} \stackrel{(1)}{=} 16 + (-5)(35 + (-2) \cdot 16)$$

$$= \underline{(-5)35 + 11 \cdot 16}$$

Thus: the inverse of $\overline{16}$ in $\mathbb{Z}_{35}$ is $\underline{\underline{11}}$.

$$\overline{16}^{-1} = \overline{11} \quad (\text{in } \mathbb{Z}_{35}).$$

$$ax \equiv c \pmod{n}$$

THEOREM 1.49. Let a and n be integers with $n \geq 2$. If a and n are relatively prime, then for every integer c the equation $\bar{a}x = \bar{c}$ can be solved for x . In fact, $x = bc$, where $ba \equiv 1 \pmod{n}$. Moreover, any two solutions are congruent modulo n .

Proof. $\gcd(a, n) = 1$

By Thm 1.45, $\bar{a}$ has an inverse, say $\bar{b}$ in $\mathbb{Z}_n$.

$$\bar{a}x = \bar{c} \Rightarrow \underbrace{\bar{b}\bar{a}}_{\bar{1}}x = \bar{b}\bar{c} \Rightarrow \boxed{x = \bar{b}\bar{c}}$$

Suppose that x_1 and x_2 are two solutions of $ax \equiv c \pmod{n}$.

$$ax_1 \equiv c \pmod{n}$$

$$ax_2 \equiv c \pmod{n} \Rightarrow c \equiv ax_2 \pmod{n}$$

$$\Rightarrow ax_1 \equiv ax_2 \pmod{n} \Rightarrow n \mid a(x_1 - x_2) \Rightarrow$$

$$\Rightarrow n \mid (x_1 - x_2) \Rightarrow x_1 \equiv x_2 \pmod{n}$$

Example 1.51. Solve the equation $\overline{16}x = \overline{9}$ in $\mathbb{Z}_{35}$.

Notice that $\overline{16}x = \overline{9}$ has a solution in $\mathbb{Z}_{35}$ since $\gcd(16, 35) = 1$.

Moreover, we know that $\overline{16}^{-1} = \overline{11}$ (in $\mathbb{Z}_{35}$).

$$\overline{16}x = \overline{9} \Rightarrow \underbrace{\overline{11} \cdot \overline{16}}_{\overline{1}} x = \overline{11} \cdot \overline{9} \Rightarrow x = \overline{99}$$

$x = \overline{99} = \overline{29}$ is a solution

Example 1.52. Solve the following system of equations in $\mathbb{Z}_{11}$:

$$\bar{5}x + \bar{8}y = \bar{2}$$

$$\bar{3}x + \bar{2}y = \bar{1}$$

$$\bar{3}x + \bar{2}y = \bar{1} \xrightarrow{\times \bar{4}} \bar{12}x + \bar{8}y = \bar{4}$$

$$\left. \begin{array}{l} \bar{5}x + \bar{8}y = \bar{2} \\ x + \bar{8}y = \bar{4} \end{array} \right\}$$

$$\bar{4}x \quad / \quad = -\bar{2} = \bar{9} \quad \text{in } \mathbb{Z}_{11}$$

We need to solve the equation $\bar{4}x = \bar{9}$ in $\mathbb{Z}_{11}$. Notice that such equation has a solution in $\mathbb{Z}_{11}$ since $\gcd(4, 11) = 1$.

We use the Euclidean Algorithm to find $\bar{4}^{-1}$ in $\mathbb{Z}_{11}$:

$$11 = 2 \cdot 4 + 3$$

$$4 = 1 \cdot 3 + \underline{1}$$

$$3 = 3 \cdot 1 + 0$$

$$\begin{aligned} 1 &= 4 + (-1) \cdot 3 = 4 + (-1)(11 + (-2) \cdot 4) \\ &= (-1) \cdot 11 + 3 \cdot 4 \end{aligned}$$

We have that $\bar{4}^{-1} = \bar{3}$ in $\mathbb{Z}_{11}$.

$$\bar{4}x = \bar{9} \xrightarrow{\times \bar{3}} \underline{x} = \bar{3} \cdot \bar{9} = \bar{27} = \underline{\bar{5}} \text{ in } \mathbb{Z}_{11}$$

$$\bar{3}x + \bar{2}y = \bar{1} \xRightarrow{\substack{\uparrow \\ x = \bar{5}}} \bar{15} + \bar{2}y = \bar{1}$$

$$\Rightarrow \bar{2}y = \bar{8} \text{ in } \mathbb{Z}_{11} \quad (\gcd(2, 11) = 1)$$

Inverse of $\bar{2}$ in $\mathbb{Z}_{11}$:

$$11 = 5 \cdot 2 + 1 \Rightarrow 1 = 11 + \underline{(-5)} \cdot 2$$

$$\bar{2}^{-1} = \bar{-5} = \bar{6} \text{ in } \mathbb{Z}_{11}$$

$$\bar{2}y = \bar{8} \Rightarrow y = \underset{\times \bar{6}}{\bar{6}} \cdot \bar{8} = \underline{\bar{48} = \bar{4}}$$

A solution for our system is:

$$x = \bar{5}, \quad y = \bar{4}$$

